Assignment - 4

Course: MST501-1: Python Programming for Statistics

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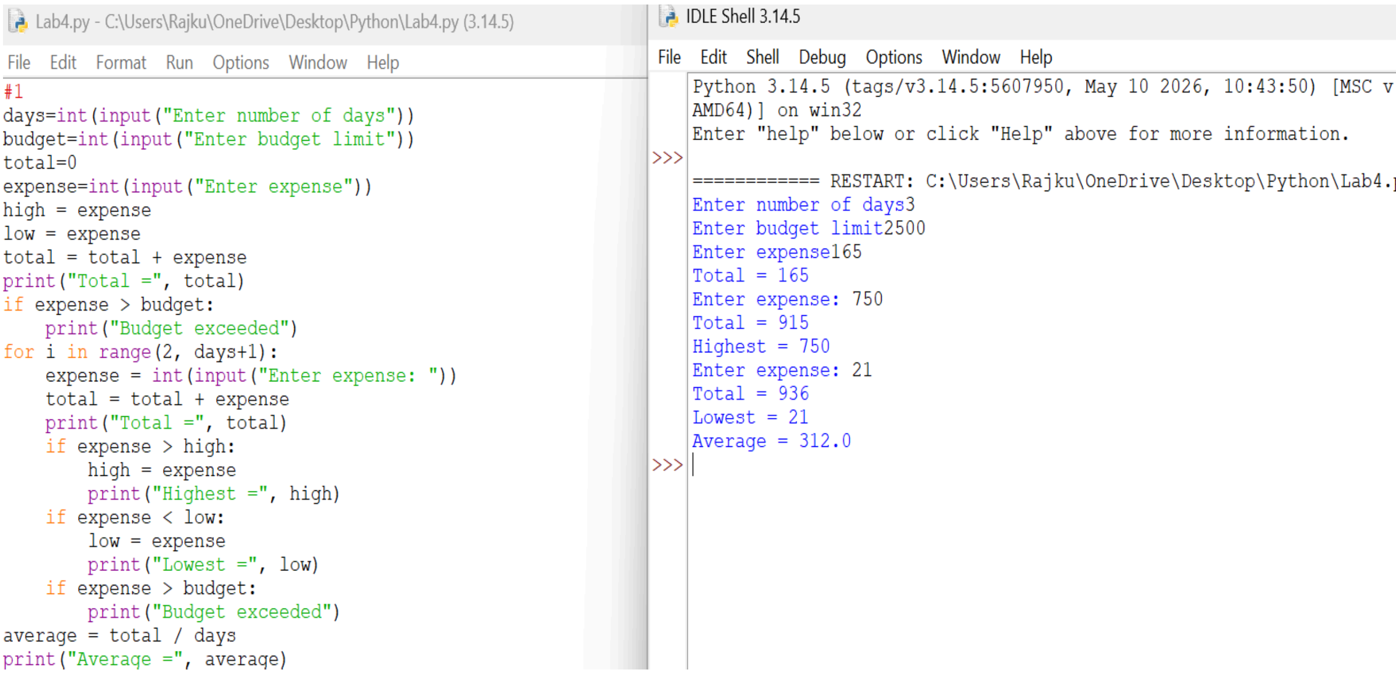
1.Develop a Python program that records a user's
daily expenses for an entire month. Using loops, calculate total spending, average daily expenditure, highest and lowest expenses, and identify days where spending exceeded a user-defined budget limit. Generate a monthly financial summary.

Ans-

```
days=int(input("Enter number of days"))
budget=int(input("Enter budget limit"))
total=0
expense=int(input("Enter expense"))
high = expense
low = expense
total = total + expense
print("Total =", total)
if expense > budget:
    print("Budget exceeded")
for i in range(2, days+1):
    expense = int(input("Enter expense"))
    total = total + expense
    print("Total =", total)
    if expense > high:
        high = expense
        print("Highest =", high)
    if expense < low:
        low = expense
        print("Lowest =", low)
    if expense > budget:
        print("Budget exceeded")
average = total / days
print("Average =", average)
```



Lab4.py - C:\Users\Rajku\OneDrive\Desktop\Python\Lab4.py (3.14.5)

File Edit Format Run Options Window Help

```
#1
days=int(input("Enter number of days"))
budget=int(input("Enter budget limit"))
total=0
expense=int(input("Enter expense"))
high = expense
low = expense
total = total + expense
print("Total =", total)
if expense > budget:
    print("Budget exceeded")
for i in range(2, days+1):
    expense = int(input("Enter expense: "))
    total = total + expense
    print("Total =", total)
    if expense > high:
        high = expense
        print("Highest =", high)
    if expense < low:
        low = expense
        print("Lowest =", low)
    if expense > budget:
        print("Budget exceeded")
average = total / days
print("Average =", average)
```

IDLE Shell 3.14.5

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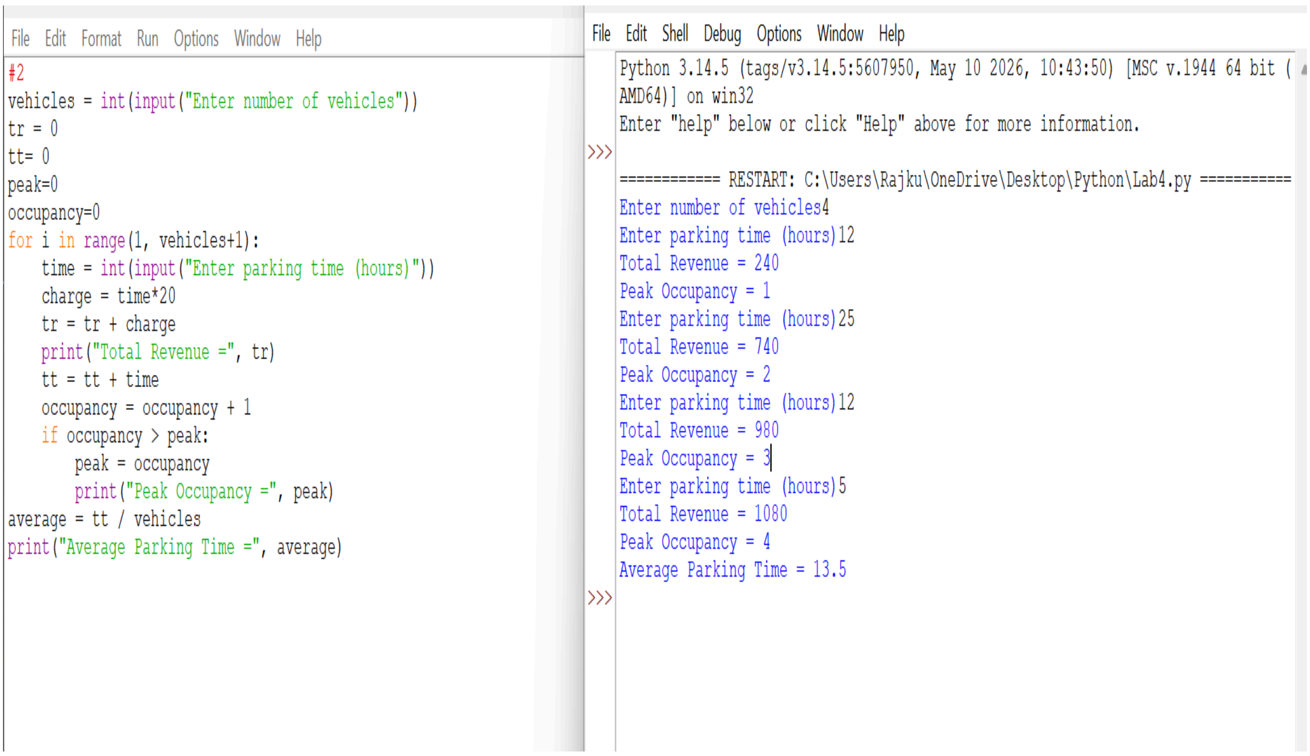
```
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============ RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Lab4.
Enter number of days3
Enter budget limit2500
Enter expense165
Total = 165
Enter expense: 750
Total = 915
Highest = 750
Enter expense: 21
Total = 936
Lowest = 21
Average = 312.0
>>>
```

2. Create a Python program that simulates a parking facility for an entire day. Using loops, process vehicle entries and exits, calculate parking charges based on duration, track occupancy levels, identify peak usage periods, and generate end-of-day statistics such as total revenue and average parking time.

Ans-
```
vehicles = int(input("Enter number of vehicles"))
tr = 0
tt= 0
peak=0
occupancy=0
for i in range(1, vehicles+1):
    time = int(input("Enter parking time (hours)"))
    charge = time*20
    tr = tr + charge
    print("Total Revenue =", tr)
    tt = tt + time
    occupancy = occupancy + 1
    if occupancy > peak:
        peak = occupancy
        print("Peak Occupancy =", peak)
average = tt / vehicles
print("Average Parking Time =", average)
```



3. Design a Python program that manages sales transactions for an online store over multiple days. Using nested loops, process customer orders, update inventory levels, calculate daily and cumulative revenue, identify best-selling products, detect low-stock items, and generate a comprehensive business performance report. The system should continue operating until the administrator chooses to terminate it.

Ans-revenue=0

```
stock=50
sold=0
choice="yes"
while choice=="yes":
    orders= int(input("Enter number of orders for the day"))
    daily= 0
    for i in range(1, orders+1):
        quantity=int(input("Enter quantity sold "))
        stock = stock - quantity
        print("Stock Left =", stock)
        sold=sold + quantity
        print("Items Sold =", sold)
        amount=quantity * 100
        daily=daily + amount
        print("Daily Revenue =", daily)
        revenue=revenue + amount
        print("Total Revenue =", revenue)
        if stock < 10:
            print("Low Stock Warning")
    choice=input("Choice")
```